Geometry : Vectors — problem 12.1.38

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A point D is chosen inside a scalene triangle ABC such that $\angle ADB = \angle ACB + 90°$ and $|AC| \cdot |BD| = |AD| \cdot |BC|$. Find

$$\frac{|AB| \cdot |CD|}{|AC| \cdot |BD|}$$ (IMO 1993).